Stationarity Considerations

Core Setup

A time series is (approximately) stationary when its statistical behaviour is time-invariant :

- its **mean** does not drift unpredictably over time,
- its **variance** is bounded and does not systematically change,
- its **autocovariance structure** depends mainly on lag, not calendar time.

In a nutshell :

- stationarity is not about precise definition but more about whether you can treat the series as coming from a **stable regime**

The purpose

- inference, forecasting, and backtesting.

Challenges

- Most macro and financial **"levels"** are shaped by **accumulation and compounding**, and by **structural changes**
  - **Macroeconomic levels** (GDP, CPI, employment, etc.) typically trend and experience regime shifts
  - **Financial prices are** often close to random walks in levels (unit-root-like)
  - **Financial returns** are often closer to stationary in mean

Fundamentals

ADF and KPSS are complementary checks on that stability

- **ADF**: asks whether the series behaves like it has a **unit root** (stochastic trend)
  - The ADF test's null is essentially :
    - **H0:** unit root exists - series has a stochastic trend - non-stationary in levels
    - **H1:** no unit root – stationary, roughly a constant or deterministic trend

- **KPSS**: asks whether the series is **stationary** (around a constant or trend)
  - KPSS is designed to confirm stationarity
    - **(H0):** the series is **stationary**.
      - Level-stationary
      - **Deterministic trend** i.e. trend is **predictable in form** (e.g., linear growth)

Special mention (Nonlinearities, regimes, heteroskedasticity**)**

- In finance, returns may be mean-stationary
- Heteroskedasticity - Volatility clusters (time-varying volatility, variance)
- Structural breaks
- Unit-root tests are not designed to model conditional variance dynamics

ADF and KPSS alignment and misalignment

ADF and KPSS are aligned scenario

- Consistency of choices about deterministic components (constant vs trend).

ADF and KPSS are NOT aligned scenario

- Data are not cleanly “stationary vs unit-root”

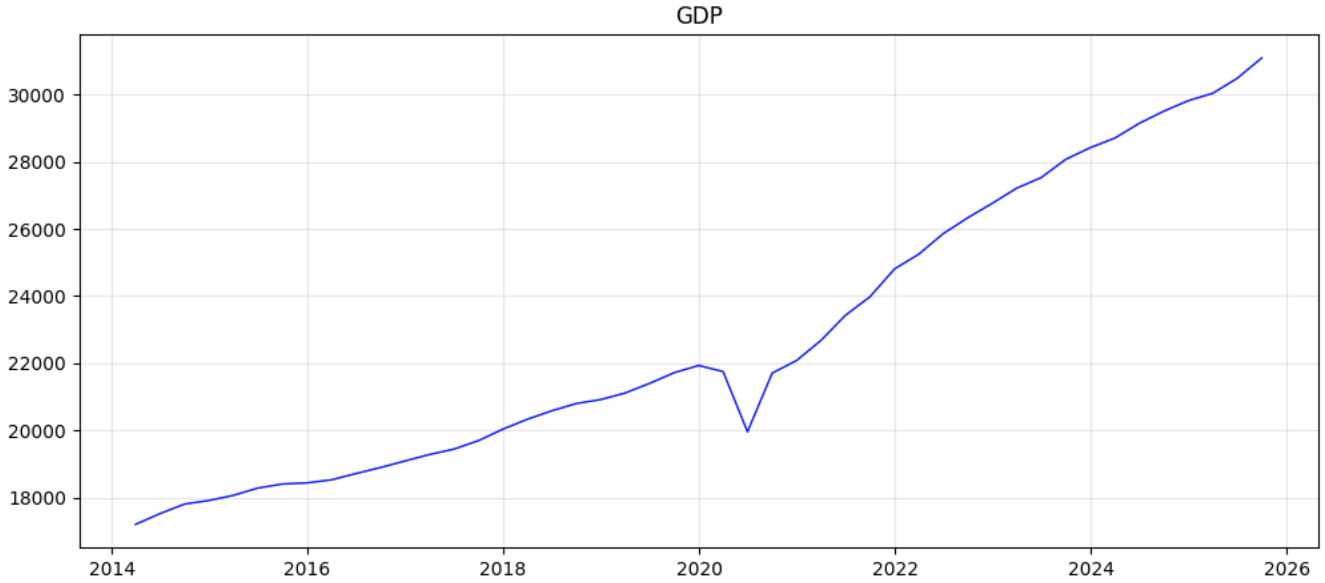
Consistency of code for testing:

Default

```
# Level-stationarity (around a constant)
adf_level  = adfuller(x, regression="c")
kpss_level = kpss(x, regression="c")

# Trend-stationarity (around a deterministic trend)
adf_trend  = adfuller(x, regression="ct")
kpss_trend = kpss(x, regression="ct")
  return go(f, seed, [])
}
```

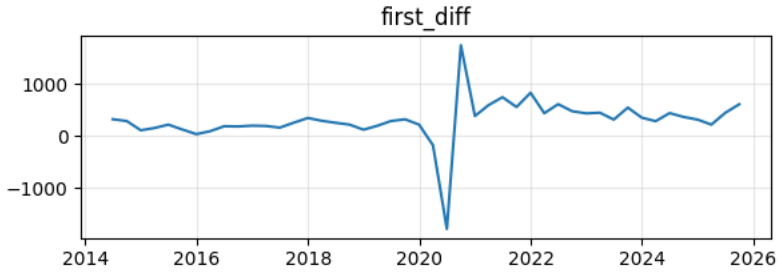
Example – Quarterly Macroeconomic Time Series

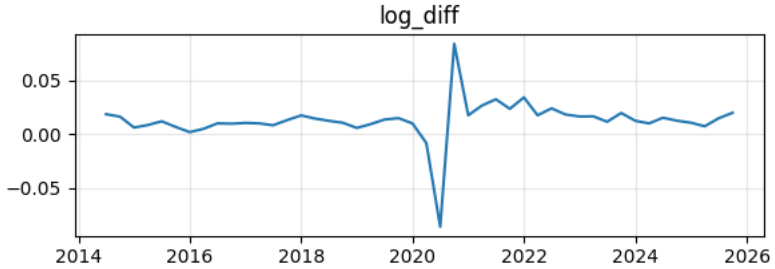


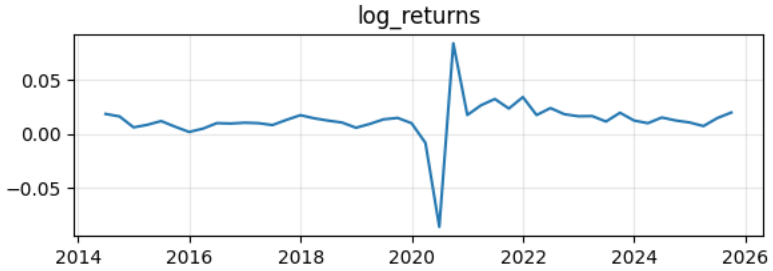
Stationarity Test Summary

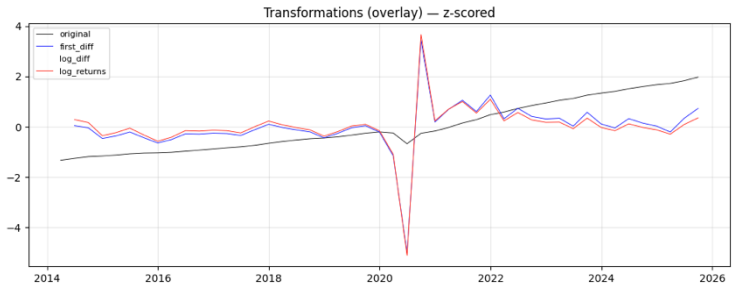
Test	Statistic	p-value	Used lags	Decision	Interpretation
ADF	1.3222	0.9967	0	FAIL TO REJECT H0	Non-stationary
KPSS	0.2393	0.01	4	REJECT H0	Non-stationary
Combined					Non-stationary

Some Transformations










```
d = (transformations["log_diff"] - transformations["log_returns"]).abs()
print("max abs diff:", d.max())
```

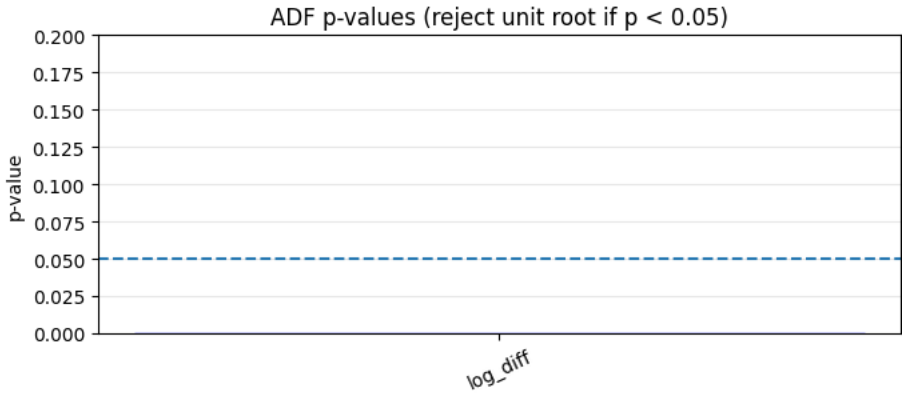
max abs diff: 1.5144135945277526e-15

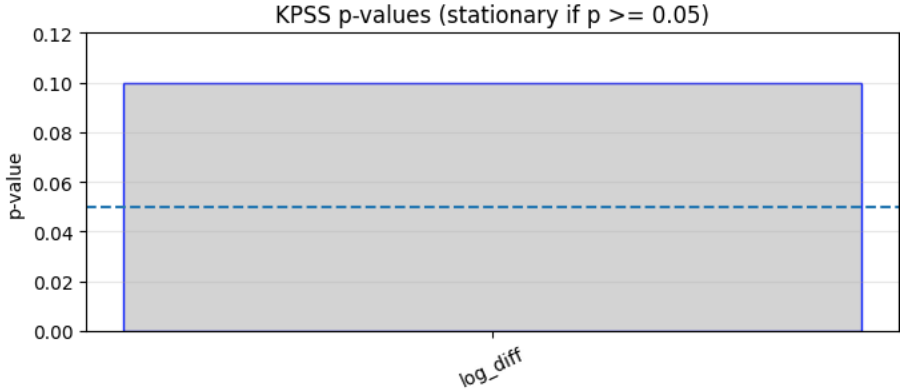
Example – Quarterly Macroeconomic Time Series

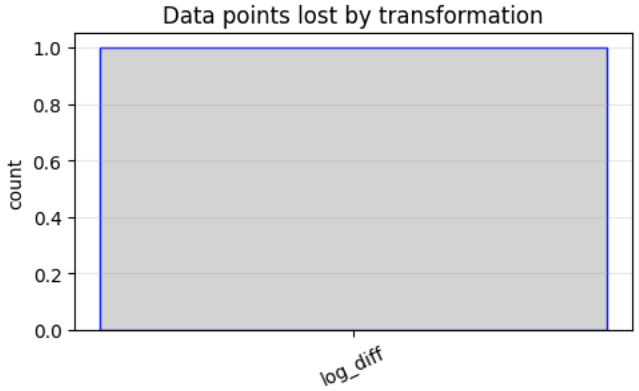
Some Transformations

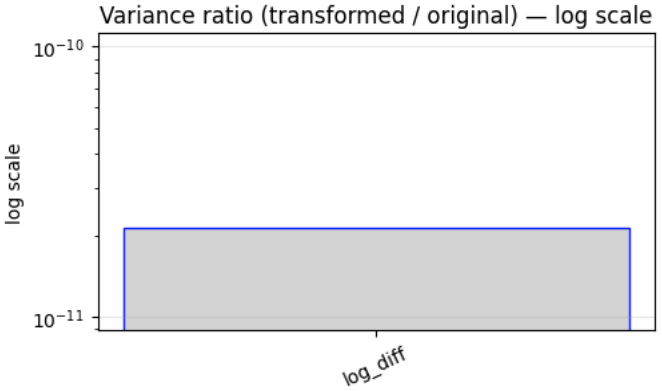
Stationarity Test Summary

Test	Statistic	p-value	Used lags	Decision	Interpretation
ADF	-7.9332	3.469e-12	0	True	Stationary
KPSS	0.1688	0.1	2	False	Stationary
Combined					Stationary

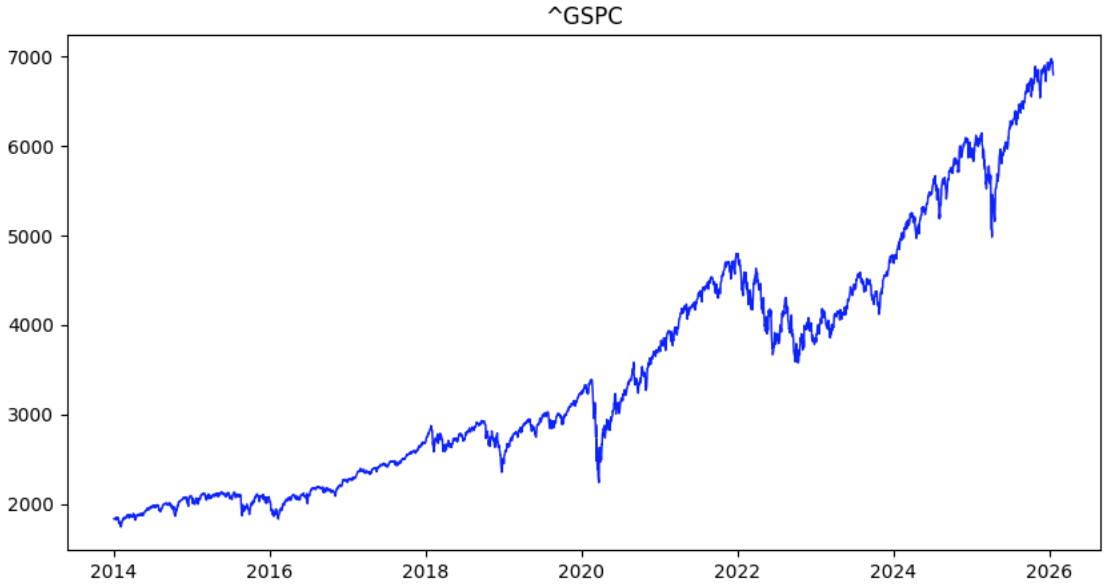








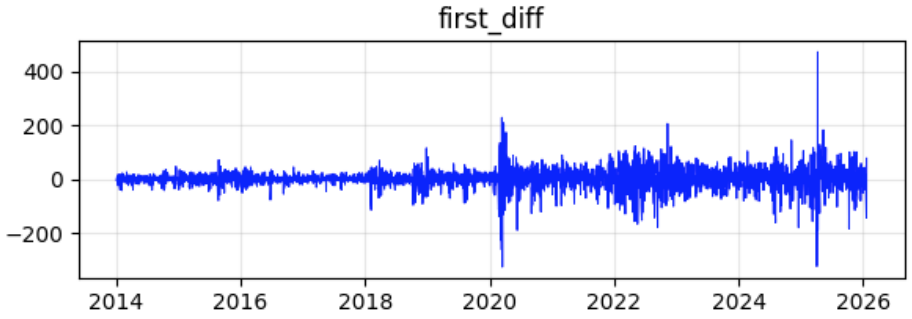
Example – Daily Markets Data

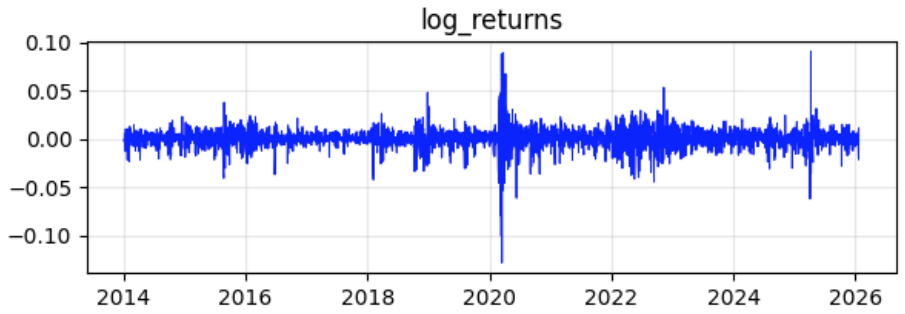


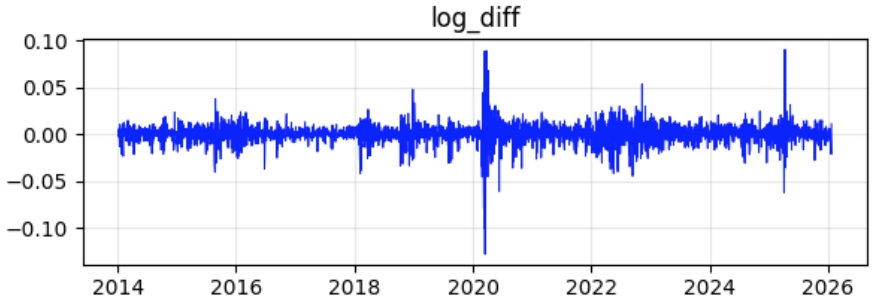
Stationarity Test Summary

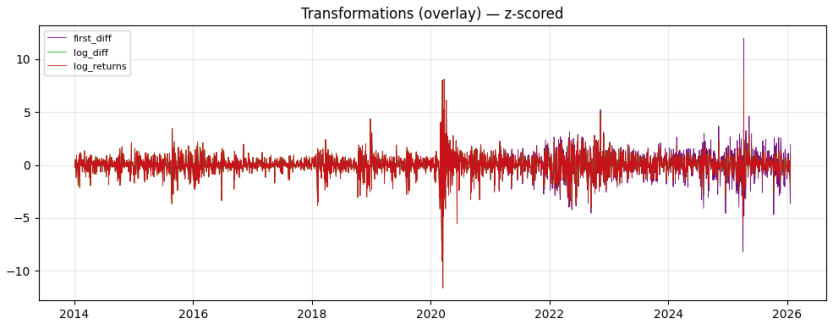
Test	Statistic	p-value	Used lags	Decision	Interpretation
ADF	-1.6989	0.7514	9	False	Non-stationary
KPSS	1.1531	0.01	32	True	Non-stationary
Combined					Non-stationary

Some Transformations





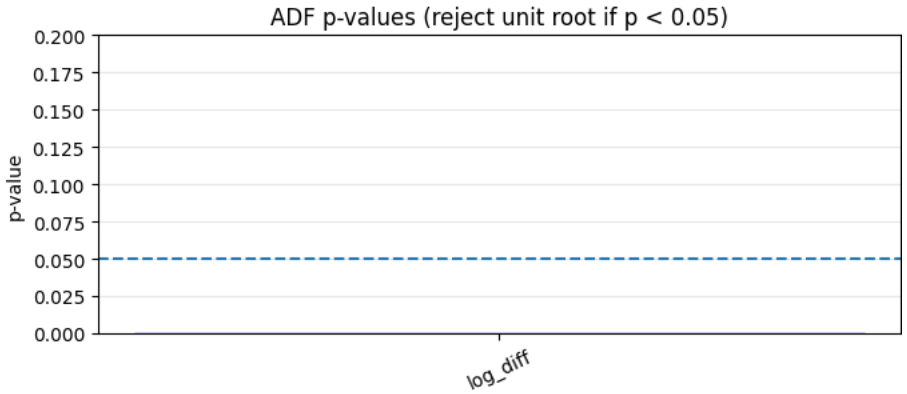


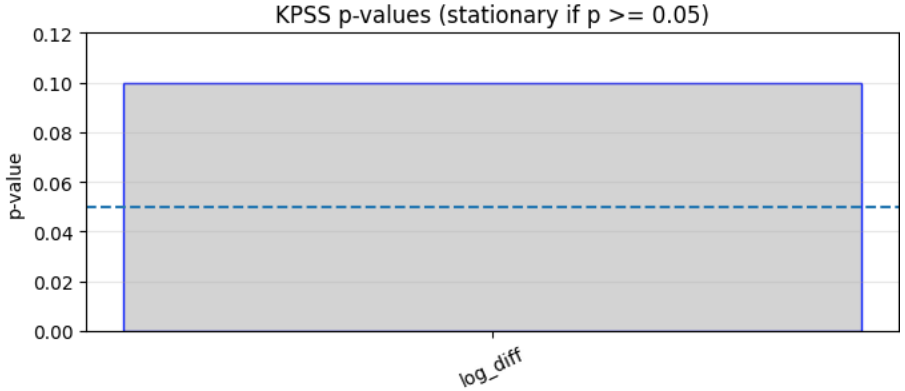


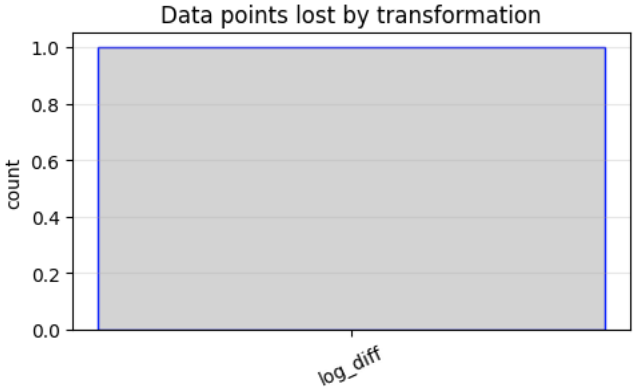
Example – Daily Markets Time Series

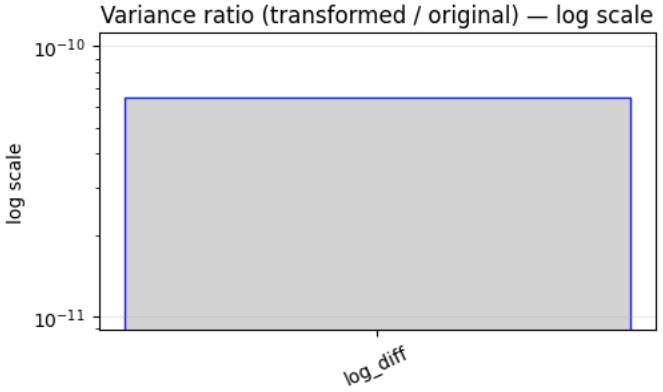
Stationarity Test Summary

Test	Statistic	p-value	Used lags	Decision	Interpretation
ADF	-17.8779	3.01e-30	8	True	Stationary
KPSS	0.0495	0.1	6	False	Stationary
Combined					Stationary









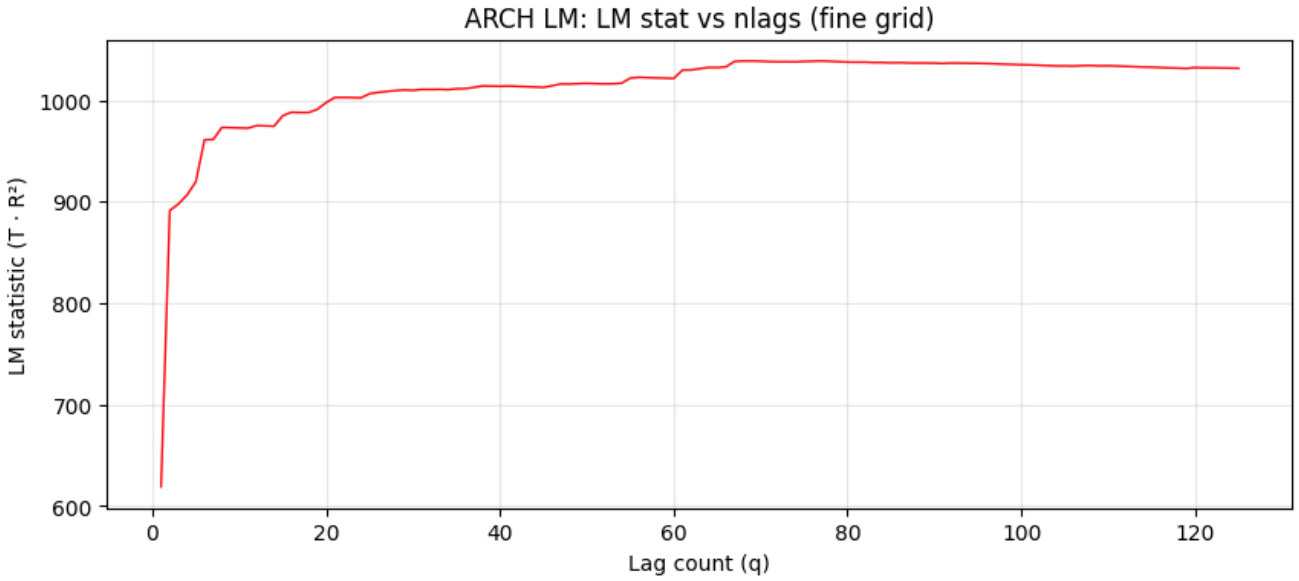
Quick Add-on Test - Engle's ARCH LM Test for Conditional Heteroskedasticity

LM – Lagrange Multiplier

Diagnose whether variance of log_diff is time-varying in a way consistent with **ARCH/GARCH effects**

ARCH: variance depends on past squared shocks:

- **Variance constant :** $Var(r_t \mid \mathcal{F}_{t-1}) = \sigma^2$
- **Variance depends on past squared shocks (ARCH)** : $Var(r_t \mid \mathcal{F}_{t-1}) = \alpha_0 + \alpha_1 r_{t-1}^2 + \cdots + \alpha_q r_{t-q}^2$



Notes

Even if returns are stationary in mean, volatility is not i.i.d.; ARCH effects persist

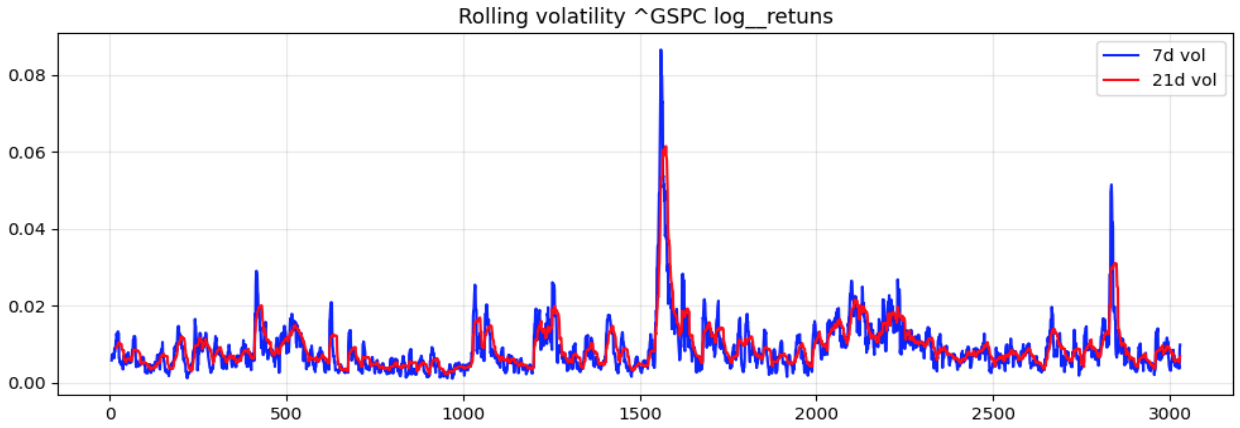
- Volatility can be **heteroskedastic** with regimes/clustering

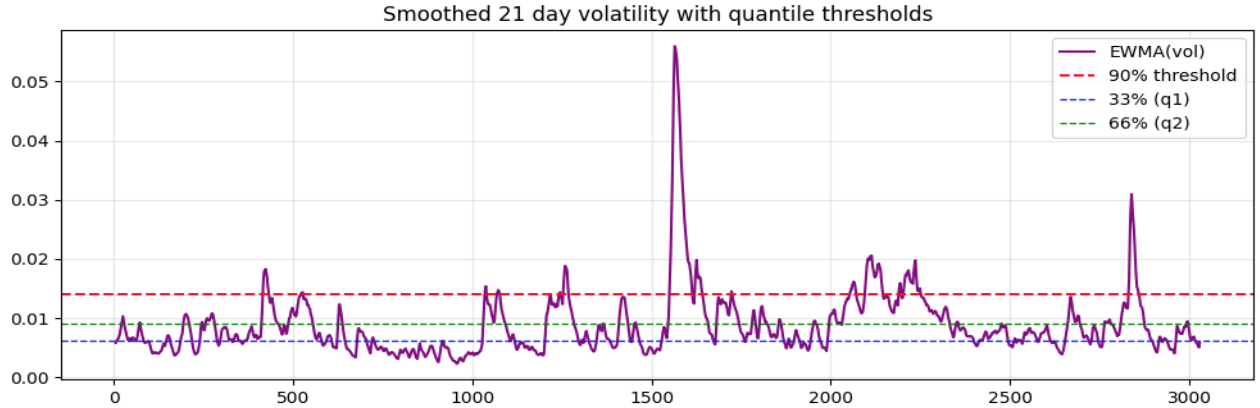
ARCH LM is a **detection test**, not an estimator of the "true" lag order!

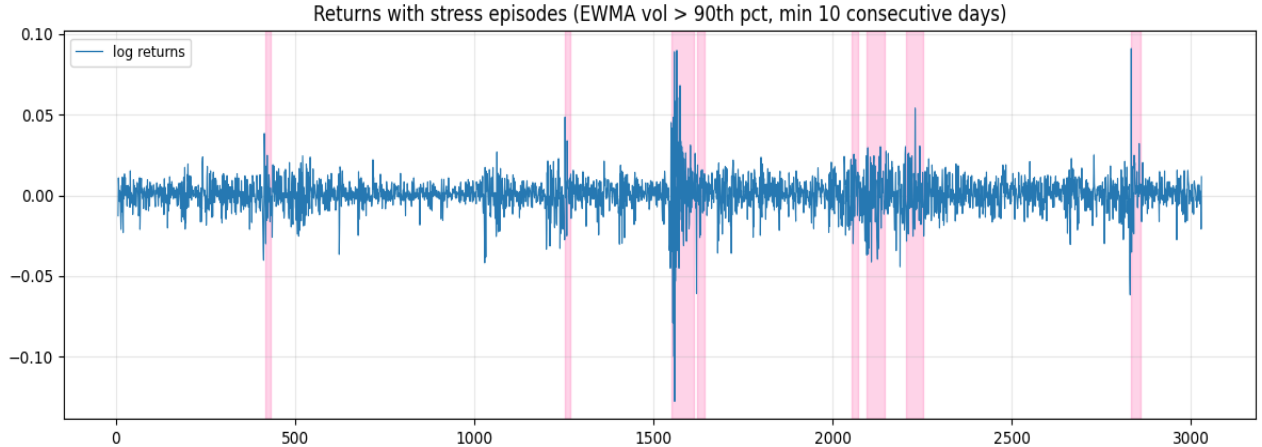
Returns maybe stationary in Mean but no i.i.d. because of variance clusters

Arch LM significant LM across many means squared – residuals Autocorrelated , volatility persistent

Simple Step Forward – Volatility State







ARCH-LM showed strong conditional heteroskedasticity (volatility clustering), so we move from:

"returns are stationary"

to

"variance is time-varying"

and operationalize this